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Mark Carson

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Summary

B.S. Computer Science graduate with hands-on experience in **geospatial systems, full-stack SWE, and data analytics**. Experienced in building **GIS-based visualization platforms** for infrastructure and telecommunications. Driven to apply scalable technology toward **social and environmental impact**.

Education

California State University, Long Beach, BS in Computer Science – Long Beach, CA, USA — Aug 2020 – Dec 2024

- GPA: **3.5/4.0**
- Study Abroad: **HAW Hamburg, Germany** (Summer 2023), GPA: 4.0/4.0, research in data analytics methodology
- Relevant Coursework: Big-Data Analytics, Machine Learning, Embedded ML, Database Design, Energy & Environment

Experience

Junior Software Engineer, Souza Consulting – Remote — July 2024 – present

- Designed and built a **full-stack RF coverage visualization platform** used by telecom engineers for multi-band signal propagation analysis through rich spatially derived information
- Implemented an interactive **OpenLayers-based geospatial mapping system**, for navigating processed GeoTIFF data from RF planning engines for click-to-query signal strength analysis for specific locations
- Built a **Node.js / Express proxy server** securely interfacing with **GeoServer WMS/WFS services** using workspace-scoped authentication for managing raster and vector geospatial data layers
- Developed **geospatial data pipelines in Python and SQL** to process coverage maps and network infrastructure datasets for a federal broadband funding program bid.

Lead Software Engineer, Mindful Metrics – Remote — Jan 2025 – present

- Led the complete development of a user-facing eCommerce management platform from the ground up, utilizing **Next.js, React, FastAPI, and PostgreSQL**
- Developed and deployed **Shopify, Inksoft, and Google Workspaces API integrations**, a React-based label management system with PDF generation, and secure authentication
- Engineered an **order fulfillment automation system** using **FastAPI and React**, processing webhooks and generating QR-coded labels, reducing manual processing time by **∼95%**

Data Scientist Intern, Divergent3D – Torrance, CA, USA — Jan 2024 – June 2024

- Automated production ticket and bill-of-materials generation using **Python and Pandas**, reducing manual effort by **87%**
- Engineered big-data pipelines and trained **machine learning models (Scikit-Learn)**, predicting part completion times with **∼80% accuracy** across 20+ manufacturing stages

Projects

Self-Hosted Infrastructure Server/Homelab — Sept 2024 – present

Personal infrastucture for hosting web applications, services, and dev. environments, ensuring reliability and security

- Deployed many services using **Proxmox**, **Docker**, and **LXC containers** for resource-efficient service isolation
- Configured secure remote access with **WireGuard VPN** and **Cloudflare Zero-Trust Tunnels** for network segmentation

Job Tracking & Networking Web Application (CSULB Capstone) — Jan 2024 – June 2024

- Served as **Project Manager** for a team of 6 using Agile methodologies and sprint-based development cycles
- Led full-stack integration using **React, Supabase, TypeScript**, Zustand, and Chart.js

Skills

Programming Languages: Python, JavaScript, SQL, Java, C++, Rust, C#

Geospatial & GIS: OpenLayers, GeoServer, GeoTIFF/Raster Processing, WMS/WFS, Geopandas, Spatial Data Pipelines

Web & Backend: React, Next.js, Node.js, Express, FastAPI, Flask, PostgreSQL, MongoDB

ML, Data, & Analytics: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, Matplotlib, Seaborn

Tools & Infrastructure: Docker, Proxmox, Git, Unix/Linux, WireGuard VPN, Cloudflare, Power BI